

The Tonic Canvas of Consciousness

A Formal Translation from Contemplative Narrative to Intrinsic Neural Rendering

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Abstract

Philosophical and contemplative traditions often describe awareness as a relatively formless field within which differentiated contents arise. Psychology and neuroscience instead describe background state, perceptual differentiation, prediction error, recurrent access, action selection, and learning. Self-Aware Networks (SAN) has proposed since 2022 that tonic oscillatory activity functions as an active “canvas of consciousness,” while phase-wave differentials (PWDs) constitute changing content within that canvas. This paper asks whether the narrative architecture of “ground, arising form, discernment, action, and return” can be translated into a mechanistic architecture without claiming that a neural model proves a religious metaphysics.

The proposed translation maps a history-conditioned tonic reference state to the field-like continuity of awareness; typed, receiver-relative deviations from that reference to arising forms; learned neural tuning to discernment; coherence-weighted recruitment to valuation and action; and reafferent / plastic updating to the return of consequences into the next reference state. Information is defined conditionally: a tonic state is not intrinsically information-free, but may have low surprisal and low entropy rate under the system’s current model, whereas a discriminable deviation may carry high information for a particular receiver. Synchrony is treated as coordination rather than consciousness itself. Both excessive synchrony and unstructured dispersion can destroy usable differentiation; conscious content is hypothesized to occupy a metastable regime in which structured differences are locally readable, recurrently available, embodied, and capable of changing future processing.

To discipline the proposed bridge, the paper combines four formal tools: a Causal Interface Sequence, a category-theoretic partial functor, a process-calculus encoding, and Gorla’s five validity criteria for encodings. A machine-checkable five-role interface model establishes an exact labeled-graph isomorphism and passes finite-interface analogues of the five criteria. Draft 2 adds a bounded source grammar, a target grammar with silent transitions, weak behavioral equivalence, semantic name substitutions, and four independent parallel contexts. The validator exhaustively checks 970 generated source terms across 3,880 term-context pairs and rejects eight adversarial mutations. The resulting claim remains deliberately bounded. The certificate proves weak operational equivalence only for the declared finite languages and contexts; a whole-tradition, full pi-calculus, empirical-neural, or ontological isomorphism is neither established nor required. The framework yields falsifiable experiments comparing predictable background activity, informative residuals, hallucinated content, minimal-content awareness, action, and memory. It also distinguishes an external computer display from intrinsic rendering: in the latter, the rendering state is part of the same embodied causal system that detects, acts, remembers, and updates itself.

Draft 3 executes the comparator and ablation benchmark that Draft 2 left open. A deterministic 768-trial synthetic design crosses two tonic contexts, two receiver signs, two content labels, and controlled noise. Tonic-only, phasic-only, pure-synchrony, and unstructured-dispersion implementations remain at chance on content; a feedforward typed classifier recovers content but fails return and memory; and complete abstract predictive and active-inference-like controls match the generic behavioral chain. The complete declared SAN chain also passes that chain, while a phase-sign intervention distinguishes its phase-typed implementation from the abstract controls. Seven single-feature ablations each break at least one preregistered observable. This is a test of declared toy implementations, not evidence of neural implementation, phenomenal identity, or SAN’s uniqueness over every predictive architecture.

Draft 4 closes the remaining local source and design gaps without changing that result. It separates six philosophical and contemplative source families instead of treating them as one doctrine; binds predictive coding and active inference to independently authored definitions while identifying the synthetic controls as local test doubles; specifies a two-stage, receiver-relative phase-measurement and closed-loop intervention protocol; and records explicit nonduplication boundaries against adjacent SAN papers. Closed-loop phase-locked human stimulation establishes feasibility for a phase-conditioned intervention, not evidence for TCT. The empirical claim remains prospective: SAN gains support only if receiver- and context-specific phase relations add held-out explanatory value and survive matched phase-blind, antiphase, sham, power, arousal, and movement controls.

Keywords: Self-Aware Networks; consciousness; tonic oscillation; phase-wave differential; information theory; nondual awareness; emptiness; neural rendering; process calculus; category theory; operational equivalence

Scope and evidence boundary

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1. The problem: similar stories at incompatible levels

Many traditions describe a recurring sequence:

1. an undifferentiated or relatively content-poor ground;
2. the arising of a form, perception, thought, or distinction;
3. recognition, appraisal, or attachment;
4. action and consequence;
5. dissolution or return into a changed background.

This sequence appears in different vocabularies across contemplative reports, philosophy of mind, phenomenology, psychology, and computational neuroscience. Similarity of sequence is not yet identity of mechanism. “Awareness,” “emptiness,” “ground,” “prediction,” “oscillation,” and “machine state” are not synonyms. A convincing bridge therefore requires more than matching prose. It must declare what is being translated, preserve causal roles and ordering, specify observables, expose mismatches, and state what result would make the translation fail.

The central proposal is that SAN supplies a candidate mechanical accounting of one bounded portion of these narratives. Its core is not that all brain activity must synchronize. It is that a living neural system maintains a recurrent reference field and constructs usable distinctions as structured deviations within that field. The field is not a passive screen and the distinction is not a picture delivered to a separate viewer. The participating cells both generate and respond to the evolving state. Their responses affect action, reafference, memory, and the next state of the field.

This paper calls that proposal the **Tonic Canvas Translation (TCT)**. TCT is a hypothesis about structural and operational correspondence. It is not a demonstration that ancient authors anticipated electrophysiology, that a contemplative doctrine reduces to one brain rhythm, or that a successful simulation would by itself establish phenomenal consciousness.

2. Historical position of the SAN claim

The proposed bridge is not a retrospective addition to SAN. The record has three date classes that must not be collapsed: owner-dated ancestry, later public Git fixation of that ancestry, and public additions that sharpened or joined the mechanism.

2.1 Owner-dated ancestry and its public boundary

- `a0348z.md` contains an internal date of May 13, 2011 and describes distributed scene “pixels,” learned statistical relations, inference, prediction, and a continuously updated multisensory picture. The surviving public Git record fixes this text on June 8, 2022 in commit `63d76b0b1b5ea1cfbddd06994ddd180c2e8ee5ea`. The 2011 date is therefore owner-dated ancestry, not an independently public 2011 timestamp.
- The initial `a0003z.md` is internally dated August 20, 2012 and describes awareness as distributed prediction and readiness, an “empty luminosity,” embodiment, and movement-oriented knowledge. It was publicly Git-fixed on June 10, 2022 in commit `7737b4f69f0de736956f6532dbbfc868072dcd9f`. The exact sentence relating the spiritual “ground of being” to tonic oscillation was added on August 4, 2022 in commit `f739086d642ecf3a04bc259f9409a3f3be0d633c` and is not backdated to 2012.
- `nlp1.md` preserves a transcript attributed to the 2017 *Neural Lace Podcast* in which a controller calculates the difference between a current neural pattern and a desired pattern and transmits that difference. It is a relevant receiver-state / PWD ancestor, but its exact 2017 public date remains source-gated until original upload metadata or an archived capture is tied to the transcript.

2.2 Public mechanistic cluster and later contemplative join

The public Git history fixes the mechanical cluster before the August contemplative synthesis:

- On June 8, 2022, the original upload `a0245z_ctpr.txt` stated that phasic spikes are evaluated relative to an oscillator’s tonic firing pattern (`a1180bfb254bb7b4329a447f71d9d6df45e78bce`).
- On June 8, 2022, `a0142z_ctpr.txt` distinguished coherence for binding and convergence from the differentiated or decoherent relations needed for distinction, and described an oscillator as creating its own reference “paper” against which incoming signals are measured as deviations (`37da575ba33f85cef4239295be161066d8d39f4c`).
- On June 8, 2022, `a0135z_ctpr.txt` described intrinsic rendering: the same embodied neural system renders, receives, predicts, and acts without a separate viewer, and the rendering predicts near-future conditions for coordinated movement (`37da575ba33f85cef4239295be161066d8d39f4c`).
- On June 11, 2022, `a0245z.md` added the exact “tonic canvas / phasic ink” formulation, together with the claim that rare phasic changes acquire meaning relative to common tonic activity (`df6e60216b38a2475a7819cfdb2baf2d3e3eb2e`).
- On August 4, 2022, `a0003z.md` proposed that the spiritual “ground of being” could describe tonic oscillation during a relatively restful condition in which incoming sensory spikes are reduced.
- On August 21, 2022, `a0004z.md` described tonic oscillatory activity as a ground of awareness from which incoming sensory content and internal patterns arise.
- On August 24, 2022, `WhitepaperNotes.md` connected collective neural response to Strogatz’s synchronized-firefly example, called common tonic activity low-information and rare phasic changes high-information, and emphasized bidirectional interaction across scales.
- The same August 24 record states in `whitepaperdraft2.md` that tonic activity is the “canvas of consciousness.” `a0004z.md` separately identifies redness as a distinct PWD relative to the tonic baseline and describes incoming high-information patterns as being transformed into a lower-information readiness state.

- On September 4, 2022, a0004z.md explicitly described differentiated and valenced forms as arising within “formless awareness,” identified there with a tonic ready state.

These notes contain hypotheses and analogies, not completed proofs. Their genealogical importance is more specific than Draft 1 reported: the public June 2022 cluster already joins receiver-relative deviation, coherence and differentiation, intrinsic rendering, prediction, action, tonic continuity, and phasic content. The August-September records then join that mechanism to the ground / form narrative and make the information-theoretic and contemplative comparison explicit.

2.3 Atom-by-year genealogy

Atom	Earliest recovered ancestry	Earliest public fixation	Later public sharpening	Claim ceiling
Distributed multisensory rendering	owner-dated 2011 a0348z.md	June 8, 2022	2022 NAPOT and whitepaper synthesis	authored ancestry; the 2011 date is not independently public
Prediction / readiness as awareness context	owner-dated 2012 a0003z.md	June 10, 2022	August 2022 tonic-ground formulation	authored ancestry plus later public mechanism
Receiver-relative difference	attributed 2017 n1p1.md transcript	exact 2017 public date unresolved	June 8, 2022 oscillator-reference formulation	2017 is source-gated; 2022 is public
Coherence for coordination versus differentiation for distinction	no earlier recovered public source in this audit	June 8, 2022 a0142z	later coherence / decoherence and PWD papers	SAN genealogy, not empirical validation
Intrinsic rendering/no separate viewer	no earlier recovered public source in this audit	June 8, 2022 a0135z	later Neural Rendering and NAPOT formulations	SAN genealogy, not proof of phenomenality
Tonic canvas / phasic ink	phase-relative precursor June 8, 2022	exact phrase June 11, 2022 a0245z	August 2022 whitepaper and redness / PWD examples	public mechanism and metaphor chronology
Ground / form narrative	owner-dated 2012 empty-luminosity discussion	exact join August 4, 2022	August-September 2022 ground, form, and valence	later join must not be backdated
Action, reafference, and update	2011 prediction / action ancestry; 2017 candidate difference-control loop	June 8, 2022 intrinsic rendering / action record	present TCT formalization	bounded functional genealogy

3. What “ground” can and cannot mean

3.1 Traditions must remain distinguishable

“Ground of being” is strongly associated with Western theology, including Paul Tillich. Advaita Vedānta’s identification of self and ultimate reality, the *Māṇḍūkya Upaniṣad*’s discussion of *turīya*, Madhyamaka accounts of emptiness, Buddhist no-self, Zen practice, and modern reports of nondual or content-minimal awareness are not a single doctrine. Treating them as interchangeable would produce a false match before formalization begins.

The present comparison therefore uses a narrow **scope lens**: descriptions in which experiential continuity is reported while ordinary differentiated content is reduced, altered, or treated as arising within a wider field. This shared narrative motif is the source object. Each tradition retains additional commitments outside the mapping.

Madhyamaka is especially important as a limit case. Emptiness is not a hidden substance or a positive cosmic field; it concerns the lack of intrinsic, independent existence and the dependence of phenomena on conditions. A physical tonic field cannot simply be labeled “Buddhist emptiness.” The more defensible correspondence is procedural: both descriptions can reject an independently existing inner observer and analyze apparent entities through relations and dependent processes. That is a partial architectural match, not an identity claim.

The source families can now be stated without collapsing them:

Source family	Source-bound motif used here	What is not imported into TCT
Tillich’s Western theology	“Ground of being” names an ontological and theological concern rather than one object among other objects (Tillich, 1952).	TCT does not identify God, being-itself, or theology with an oscillatory baseline.
<i>Mandukya Upanisad</i>	A “fourth” is distinguished from waking, dreaming, and deep sleep and is not simply another ordinary content state (Olivelle, 1998).	TCT does not establish <i>turiya</i> , Brahman, or the metaphysics of the text.
Advaita Vedanta	Later Advaita philosophy develops a nondual account of self and ultimate reality (Deutsch, 1969).	A history-conditioned neural reference is not Atman or Brahman, and neural modeling does not adjudicate Advaita.
Madhyamaka	Nagarjuna ties emptiness to dependent arising and treats emptiness itself as dependently designated rather than an ultimate substrate (Siderits & Katsura, 2013, ch. 24).	Emptiness is not renamed as a neural field, substance, or background signal.
Soto Zen through Dogen	Practice language decenters a fixed inner self by describing the self in relation to the wider field of phenomena (Dogen, 2013, “Genjokoan”).	This paper does not turn Zen practice, realization, or nonattachment into one neural mechanism.
Contemporary content-minimal awareness research	First-person reports can be paired with EEG and fMRI measures under a declared neurophenomenological protocol (Varela, 1996; Winter et al., 2020).	A report and its correlates do not prove a universal content-free neural state or TCT.

The common object is therefore not a doctrine. It is a deliberately thin relational pattern: relatively stable continuity, differentiated arising, changed access or appraisal, action or report, and a consequential return. Formal correspondence is evaluated only for that pattern. The source-specific commitments in the rightmost column are outside the translation and cannot be recovered by relabeling TCT variables.

3.2 A neural “ground” is active and conditional

In TCT, the neural ground is not silence. It is a history-conditioned distribution over possible near-future states. Let the embodied neural state be x_t , its accessible history be h_t , and contextual variables be z_t . The system’s tonic reference can be represented as

$$q_t(x_{t+1} \mid h_t, z_t). \quad (1)$$

The corresponding expected state is

$$\hat{x}_{t+1} = \mathbb{E}_{q_t}[x_{t+1} \mid h_t, z_t]. \quad (2)$$

This reference is “formless” only in a restricted phenomenological sense: it is not yet the particular differentiated content under study. Physically it remains structured, metabolically maintained, and causally consequential.

4. Conditional information and the tonic canvas

4.1 Common does not mean meaningless

SAN’s 2022 shorthand called common tonic patterns low-information and rare phasic deviations high-information. The precise information-theoretic formulation is receiver- and model-relative. For an event x under reference model q_t , surprisal is

$$I_t(x) = -\log_2 q_t(x \mid h_t, z_t). \quad (3)$$

A frequently predicted tonic event can have low surprisal under q_t , while a rare deviation has higher surprisal. This does not mean the tonic state contains no physical structure or no information under every code. It means the event adds relatively little new information to a receiver already adapted to that distribution.

The entropy of the next-state distribution is

$$H_t(X) = -\sum_x q_t(x \mid h_t, z_t) \log_2 q_t(x \mid h_t, z_t). \quad (4)$$

For continuous, nonstationary neural activity, an entropy rate or conditional predictive-information measure is generally more appropriate than a single global entropy (Shannon, 1948). TCT therefore predicts not “tonic equals low entropy” universally, but a local contrast between predictable reference activity and deviations that a learned receiver can discriminate and use.

4.2 Phase-wave differential as typed residual

For receiver (r), define a PWD-like residual

$$d_r(t) = T_r[x(t)] - T_r[\hat{x}(t)], \quad (5)$$

where T_r is the receiver’s learned transformation: its dendritic geometry, synaptic weights, membrane state, channel kinetics, and network context. The same physical input can therefore produce different informational consequences for different receivers.

A useful novelty score is a log-likelihood ratio between an event model p_r and the tonic reference:

$$\Lambda_r(t) = \log \frac{p_r(T_r[x(t)] \mid h_t, z_t)}{q_r(T_r[x(t)] \mid h_t, z_t)}. \quad (6)$$

Neither a large residual nor high surprisal alone is sufficient for conscious content. Random noise can be surprising without being interpretable. TCT requires that a deviation be structured, receiver-readable, recurrently available, and integrated into the organism’s ongoing control loop.

5. Synchrony is coordination, not the content of consciousness

Strogatz’s fireflies and clocks illustrate how coupled oscillators produce collective timing. Let $\theta_j(t)$ be the phase of unit j . The Kuramoto order parameter is

$$R(t)e^{i\Psi(t)} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)}, \quad 0 \leq R(t) \leq 1. \quad (7)$$

High (R) describes phase concentration; low (R) describes phase dispersion. Neither value, by itself, identifies consciousness. Neural systems can exhibit local synchronization that supports communication, excessive pathological synchronization, structured phase gradients, metastable coalitions, or unstructured noise.

The key distinction is between **dispersion** and **informative differentiation**. Define a structured-difference term

$$D_{\text{str}}(t) = I(d(t); Y_{t:t+\tau} \mid h_t, z_t), \quad (8)$$

where Y contains future discriminations, actions, reports, or memory updates. A deviation is valuable when it predicts or controls downstream outcomes, not merely when phases are scattered.

This leads to a non-monotonic hypothesis. Let $C(t)$ be coherence, $D_{\text{str}}(t)$ usable differentiation, $A(t)$ recurrent availability, and $B(t)$ embodiment / self-update. A candidate content score is

$$\Phi_{\text{TCT}}(t) = g(C(t)) D_{\text{str}}(t) A(t) B(t), \quad (9)$$

where $g(C)$ is small under both globally rigid locking and uncontrolled incoherence, and larger in a metastable middle regime. This is not an established consciousness index. It is a testable organizing quantity.

The model explains why gamma synchrony alone was never sufficient. Gamma-band timing may route or gate information, but the informational content lies in differences that are readable against a reference and made available to the system. Face perception has been reported with transient long-distance gamma synchronization followed by active desynchronization between perception and motor response (Rodriguez et al., 1999). Conversely, intracranial postictal recordings and cross-species anesthesia / sleep results show that strong phase coupling or hyper-correlated gamma can accompany unconsciousness (Pockett & Holmes, 2009; Bola et al., 2017). These findings are compatible with the proposed distinction; they do not by themselves prove TCT.

6. Form, hallucination, and source attribution

The tonic canvas metaphor becomes scientifically useful only if it distinguishes externally driven perception, imagery, dream, and hallucination without requiring a separate viewer.

Let s_t denote sensory evidence, m_t memory and internal-model state, and u_t interoceptive / control state. A rendered content state is proposed as

$$y_t = \mathcal{R}(s_t, m_t, u_t, x_t), \quad (10)$$

with source-attribution variable

$$\alpha_t = P(\text{external cause} \mid s_t, m_t, u_t, y_t). \quad (11)$$

In ordinary perception, external evidence constrains y_t and source attribution is often high. In imagery or dreaming, m_t can dominate while the system retains or loses awareness of the internal source. In hallucination, internally generated or weakly constrained content may be assigned an external cause. In content-minimal awareness, differentiated y_t may be reduced while continuity, arousal, or self-related process remains.

Oliver Sacks’s clinical narratives are relevant because they show the diversity of hallucinated form and the importance of lesion, sensory loss, network state, and source attribution (Sacks, 2012). They do not establish a universal tonic canvas. Likewise, a philosophical or contemplative report of “clear,” “formless,” or attribute-poor awareness is phenomenological evidence about a reportable state, not direct evidence for one oscillatory substrate. Neurophenomenology supplies one disciplined way to coordinate first-person and physiological evidence without collapsing them (Varela, 1996), and the content-free-awareness case study by Winter et al. (2020) illustrates both the promise and the severe single-case limits of that program.

The user-attributed Ken Wilber phrase “a color that was no color” is therefore treated here as a research target rather than a verified quotation. If a primary recording or publication is recovered, its exact context should be analyzed. Until then, it cannot carry a clinical or historical inference.

7. Intrinsic rendering versus an external display

A computer framebuffer can encode a scene while remaining an artifact for an external viewer. SAN’s stronger claim is that biological neural rendering is **intrinsic**: the rendered state participates in the same system’s detection, action, memory, maintenance, and future transformation.

Define a causal-closure profile for candidate rendering state y_t :

$$K_{\text{int}}(y_t) = w_1 I(y_t; x_{t+1}) + w_2 I(y_t; a_{t:t+\tau}) + w_3 I(y_t; m_{t+\tau}) + w_4 I(y_t; u_{t+\tau}), \quad (12)$$

where a is action, m memory, and u organismic control. The terms should be estimated under perturbation, not mere correlation. A high value would show that the candidate rendering is embedded in self-modifying control. It would not, alone, prove phenomenality.

The operational distinction is therefore:

- **extrinsic display**: the representation primarily affects an outside observer unless additional machinery reads it;
- **intrinsic rendering candidate**: the representation is generated, detected, acted through, remembered, and updated by one recurrent embodied system.

This distinction does not imply that silicon machines are necessarily nonconscious. A recurrent, embodied, self-maintaining machine could satisfy some or all of the operational conditions. TCT supplies tests rather than a substrate prohibition.

8. Causal Interface Sequence: the shared narrative architecture

The patent materials define a Causal Interface Sequence (CIS) as an ordered directed chain of operational roles rather than an unordered list of similar terms. For this paper, the phenomenological and SAN sequences are:

$$\text{Ground} \rightarrow \text{Arising Form} \rightarrow \text{Discernment/Valuation} \rightarrow \text{Action} \rightarrow \text{Consequence/Return}, \quad (13)$$

$$q_t \rightarrow d_r(t) \rightarrow T_r\text{-selection / gain} \rightarrow a_t \rightarrow (s_{t+1}, m_{t+1}, q_{t+1}). \quad (14)$$

The proposed atom map is:

Narrative role	SAN operational atom	Preserved relation	Major remainder
Ground or relative formlessness	Tonic, history-conditioned reference q_t	supplies continuity and a contrast class	no metaphysical ultimacy follows
Arising form	Receiver-relative PWD $d_r(t)$	difference appears within an ongoing field	not every deviation is phenomenal
Discernment	Learned tuning T_r , recurrent access	form becomes discriminable and usable	conceptual recognition may need language and memory
Valuation or attachment	gain, salience, action-policy weighting	some forms acquire greater control	ethical / spiritual meanings exceed gain control
Action	motor / autonomic / cognitive output a_t	differentiation changes the world or internal state	contemplative non-action is not simple motor silence
Consequence and return	reaffERENCE, plasticity, updated q_{t+1}	effects become conditions of future processing	karma is not reduced to one learning rule
No separate inner witness	distributed recurrent causal closure	observer work is performed by the organized process	phenomenal subjectivity remains an open explanatory problem

This is a genuine narrative-architecture match under the declared lens because roles, direction, and feedback are preserved. It is not yet a proof of mechanism across traditions.

9. Category-theoretic statement of the bridge

Let $\mathcal{P}$ be a category whose objects are operationalized phenomenological states and whose morphisms are admissible transitions among them. Let $\mathcal{S}$ be a category whose objects are SAN state descriptions and whose morphisms are neural / control transitions. A translation is a mapping

$$F : \mathcal{P}_{\text{scope}} \rightarrow \mathcal{S}. \quad (15)$$

For (F) to be a functor on the bounded subcategory, it must preserve identity and composition:

$$F(\text{id}_p) = \text{id}_{F(p)}, \quad F(g \circ f) = F(g) \circ F(f). \quad (16)$$

The second condition is the crucial one. Translating “form arises, is appraised, and changes action” must yield the same target transition as translating each component and composing them. If the mapping works only by redescribing the whole story after the fact, it is not compositional evidence.

A full categorical isomorphism of the traditions would require an inverse functor recovering every relevant structure without loss. That is implausibly strong: contemplative traditions contain ethical, soteriological, metaphysical, and linguistic structure absent from this neural model, while SAN contains biophysical details absent from the narratives. The machine-checkable model accompanying this paper does establish an

exact isomorphism between two deliberately finite five-state labeled graphs. The appropriate general claim remains a **partial or lax functor**, potentially faithful over the selected CIS but neither full nor essentially surjective over the entire traditions.

This formal limitation is a strength. It identifies exactly what the machine account claims to preserve and what it leaves untouched.

10. Process-calculus encoding

Category theory describes compositional structure (Baez & Fong, 2018); a process calculus makes interaction and observable behavior explicit (Milner, 1999). Define source language L_P for the bounded phenomenological narrative and target language L_S for SAN interactions. In a π -calculus-style target, a receiver process can be sketched as

$$\text{Cell}_i(s_i) \triangleq \text{in}_i(v).(\tau.\text{Update}_i(s_i, v) \mid \overline{\text{out}_i}\langle w_i \rangle).\text{Cell}_i(s'_i), \quad (17)$$

where reception, internal transformation, emission, and state update recur. Coupling channels can be created, strengthened, weakened, or rerouted. A population implements the tonic reference through recurring coupled interactions; an event produces a trace of typed deviations; action and sensory return close the loop.

Let the translation be $\mathcal{E} : L_P \rightarrow L_S$. Observable barbs are deliberately limited to events that can be measured: successful content discrimination, source attribution, report, action, and memory update. Let $\checkmark$ denote the declared success condition that a form becomes discriminable and causally usable while continuity of the embodied process remains.

Draft 2 makes a bounded fragment executable. Its source terms are

$$P ::= 0 \mid \checkmark \mid \alpha.P \mid P \oplus P \mid P \parallel P, \quad (18)$$

where α ranges over typed `arise(content)`, `discern(value)`, `act(action)`, `return(memory)`, `settle(baseline)`, and a declared fading event. The target fragment is

$$T ::= 0 \mid \checkmark \mid \tau.T \mid \beta.T \mid T \oplus T \mid T \parallel T. \quad (19)$$

The encoding maps every source-visible label injectively to a target-visible label, preserves choice and parallel composition, and inserts one finite internal τ transition after each mapped visible action. Weak visible behavior is therefore evaluated through the closure

$$T \xRightarrow{\ell} T' \iff T(\rightarrow_\tau)^* \xrightarrow{\ell} (\rightarrow_\tau)^* T'. \quad (20)$$

Let $\mathcal{G}_2$ be the exhaustively generated grammar through constructor depth two, $\mathcal{F}_5$ the complete declared five-stage trace family, and $\mathcal{C} = \{I, O, D, M\}$ the identity, observer, distractor, and memory-monitor contexts. The machine-checked result is

$$\forall P \in (\mathcal{G}_2 \cup \mathcal{F}_5), \forall C \in \mathcal{C} : C[P] \approx_{\text{wb}} C[\mathcal{E}(P)], \quad (21)$$

where $\approx_{\text{wb}}$ is the greatest bounded weak-bisimulation relation after applying the declared source-to-target label map. Success is evaluated through τ -closure. This certificate covers 970 unique source terms and

3,880 term-context pairs. It does not include unbounded recursion, synchronization handshakes, channel restriction, or name passing.

The π -calculus vocabulary is an analytical overlay. It is not native vocabulary of Zen, Advaita, Madhyamaka, psychology, or the original SAN notes. Its value is that it prevents a metaphorical match from masquerading as a behavioral proof.

11. Gorla audit: what would count as a valid encoding?

Gorla (2010) proposed five criteria for evaluating encodings between process calculi. Adapted carefully, they define the proof obligations for TCT.

11.1 Compositionality

The translation of a composite narrative must be constructed from translations of its parts. “Ground + arising form + response” cannot receive a bespoke neural story that is unavailable to its components.

Draft 2 status: bounded-grammar pass. The encoding preserves zero, success, visible prefix, binary choice, and binary parallel composition across all 970 generated source terms. Draft 1’s edge / path result remains the rollback baseline. Structural induction for an unbounded extensible grammar remains open.

11.2 Name invariance

Renaming a source symbol should induce a corresponding renaming in the target rather than changing the mechanism. Synonyms such as “background,” “canvas,” or “relative formlessness” should not manufacture a match. Conversely, doctrinally different terms must not be collapsed merely because their English translations resemble one another.

Draft 2 status: bounded semantic-renaming pass. The validator applies injective content, value, action, and memory substitutions such as red / blue, positive / negative, approach / avoid, and reinforce / revise, then verifies that substitution before encoding agrees with substitution after encoding. This is stronger than changing ignored display strings. Cross-tradition synonym, false-friend, and doctrinal-non-equivalence tests remain open.

11.3 Operational correspondence

Completeness asks whether each admissible source transition can be represented by the target. Soundness asks whether target transitions claimed as translations correspond to an admissible source transition.

Draft 2 status: bounded weak operational completeness and soundness pass for 3,880 term-context pairs. Every declared source-visible move is weakly matched by the target, and every target-visible move is traced through the inverse label map to a source move. Full pi-calculus and empirical correspondence proofs remain open.

11.4 Divergence reflection

The translation must not introduce endless or pathological internal activity absent from the source. This matters because a recurrent neural network can oscillate indefinitely while producing no usable content.

Draft 2 status: bounded weak divergence-reflection pass. The baseline target inserts finite silent steps but no reachable silent strongly connected component. A target-only tau-livelock mutation is rejected. Biophysical stability conditions, stopping criteria, and pathological neural attractors remain open.

11.5 Success sensitiveness

The source reaches its declared success state if and only if the translation does. Here, success is not spiritual liberation. It is the narrow operational event $\checkmark$: discriminable content becomes available to report, action, or memory while system continuity remains.

Draft 2 status: bounded weak-success pass in all four contexts. Success is compared through silent-transition closure, and a premature-success mutation is rejected. Empirical success sensitivity remains unproved; Section 13 specifies initial tests.

11.6 Machine-checkable bounded certificate

The accompanying model defines five source states, five target states, five directed transitions, role labels, observable sets, success states, and explicit state / action maps. Its validator checks totality, bijection, role and observable preservation, identity and composition, finite-interface versions of all five Gorla criteria, and exact labeled-graph isomorphism. Seven mutation tests break the state bijection, completeness, soundness, observable preservation, success preservation, or divergence reflection and confirm that the validator fails closed.

The certificate therefore proves an exact isomorphism of the **declared five-role interface**. It does **not** certify a Gorla-valid encoding of an entire contemplative tradition into a full SAN process calculus. That larger result still requires typed grammars, general reduction semantics, contexts, and empirical interpretation.

11.7 Bounded weak process-encoding certificate

Draft 2 preserves the five-role graph certificate and adds an independently executable behavioral layer. The source generator enumerates 970 normalized terms from the declared grammar through constructor depth two and adds the complete five-stage trace family. Each term is tested inside four interleaving contexts: identity, observer, distractor, and memory monitor. The target is allowed one finite internal transition after each mapped visible event, so equivalence is evaluated weakly rather than by requiring identical immediate states.

The final run checks all 3,880 term-context pairs and reports no unmatched source move, unmatched target move, premature success, or target-only silent divergence. It also verifies bounded compositionality and semantic name invariance. Eight mutations are rejected: missing action, spurious action, target-only tau divergence, semantic name collision, serialized parallel composition, premature success, incorrect memory attribution, and target deadlock.

The first behavioral run was deliberately retained as a failure artifact. It compared success only in the immediate state and therefore rejected a target whose success was reachable after a single finite silent step. Draft 2 corrected the verifier to compare weak success through tau-closure. The divergence test was not weakened: target-only tau cycles still fail. Preserving both receipts distinguishes a corrected semantic definition from an erased negative result.

The certified claim is therefore: **for the declared finite grammar, complete five-stage trace family, semantic substitutions, and four contexts, the translation preserves weak observable behavior, divergence, and success.** It is not a theorem about unbounded recursion, name passing, all contemplative narratives, actual neural tissue, or phenomenal identity.

11.8 Bounded tonic-phasic comparator and ablation certificate

Consider the ordinary problem of recognizing the same event while the background changes and while two receivers encode the event with opposite local signs. A detector that reads only the background misses the event. A detector that reads only the raw event channel mistakes background displacement for content. A detector that ignores receiver type assigns opposite meanings to physically inverted encodings. Even a correct classifier does not yet show that its representation is used by the same system for action, sensory return, and memory.

The inherited scientific vocabulary is straightforward: context normalization, signed coding, intervention, generalization, refference, and memory update. In plain terms, the model must subtract the active context, interpret the remaining change according to the receiving channel, use that difference to select an action, receive the consequence, and let the consequence alter a retained state. Only after those operations are explicit does this paper call their joined implementation the **bounded tonic-phasic comparator certificate**.

The synthetic test family crosses tonic context $b_c \in \{-3, +3\}$, receiver sign $\sigma_r \in \{-1, +1\}$, content $y \in \{-1, +1\}$, and Gaussian noise:

$$x = b_c + \sigma_r y + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.2^2). \quad (22)$$

The complete typed decoder is

$$\hat{y} = \text{sign}[\sigma_r(x - b_c)]. \quad (23)$$

The benchmark does not judge a mechanism from classification alone. For mechanism M , it records the score vector

$$\mathbf{s}(M) = (\text{Acc}_{\text{content}}, \text{Acc}_{\text{action}}, \Delta_{\text{self}}, \Delta_{\text{return}}, \Delta_{\text{memory}}, \Delta_{\text{phase}}). \quad (24)$$

Here Δ_{self} is the action-accuracy drop when the internal representation is silenced; Δ_{return} and Δ_{memory} compare forced-correct with forced-wrong actions; and Δ_{phase} is the action-accuracy drop after receiver phase signs are inverted. A generic behavioral pass and a phase-sensitive mechanism pass are deliberately separate:

$$\begin{aligned} G(M) &= \bigwedge_{k \in \{\text{content}, \text{action}, \text{self}, \text{return}, \text{memory}\}} s_k(M) \geq \tau_k, \\ P(M) &= G(M) \wedge \text{phase-typed}(M) \wedge \Delta_{\text{phase}} \geq 0.4. \end{aligned} \quad (25)$$

The generated suite evaluates nine declared toy mechanisms and seven single-feature ablations over 768 balanced trials. The complete SAN chain passes G and P . Each ablation breaks at least one protected observable. Crucially, the abstract predictive controller and active-inference-like controller also pass G . The suite therefore rejects a uniqueness claim based on behavior alone. Its positive result is narrower: within this declared family, tonic context and typed phasic difference are jointly required for context- and receiver-invariant content recovery, while internal phase perturbation is a candidate discriminator between a phase-typed implementation and behaviorally equivalent abstract controls.

The certificate has three limits. First, the mechanisms are declared test doubles, not exhaustive representatives of the named theory classes. Second, the phase flip is a synthetic internal intervention, not a neural experiment. Third, passing P does not establish consciousness. The executable value is that the

suite separates generic feedback success, phase-sensitive implementation, and phenomenality rather than allowing one to stand in for another.

11.9 Independently authored comparator definitions and the implementation boundary

The ordinary problem is that a locally named “predictive controller” can be made to pass a task without faithfully implementing a published predictive theory. A behavioral score then compares test doubles, not whole research programs. The published mechanisms must be defined before the local controls are interpreted.

Rao and Ballard’s hierarchical predictive-coding proposal assigns predictions of lower-level neural activity to feedback pathways and residual errors to feedforward pathways (Rao & Ballard, 1999). Friston and colleagues’ active-inference process theory derives belief updating and action selection from a generative model through variational free-energy minimization and belief propagation (Friston et al., 2017). These are independently authored definitions. They are not interchangeable with one another, with all predictive-processing theories, or with the minimal switches in the Draft 3 suite.

Comparator label in Draft 3	Independently authored source definition	What the local test double instantiates	What remains unavailable in this package
Abstract predictive controller	Hierarchical top-down prediction and feedforward residual error in Rao and Ballard (1999)	Context subtraction, typed content recovery, action, return, and memory	The source model’s hierarchy, learned generative weights, neural receptive-field predictions, and an author-supplied implementation adapted to this paper’s interface
Active-inference-like controller	Belief propagation and action selection under a generative model minimizing variational free energy in Friston et al. (2017)	The same generic perception-action-return score vector, without a phase-typed variable	Variational message passing, expected free energy, policy inference, precision dynamics, and an author-supplied implementation adapted to this paper’s interface
SAN tonic-phasic chain	The declared TCT signature in this paper	Tonic context, receiver-relative typing, intrinsic use, action, return, memory, and synthetic phase sensitivity	Empirical neural identification and comparison with independently implemented, capacity-matched alternatives

No independently authored reference implementation conforming to the fixed this paper synthetic inputs, interventions, and outputs was recovered or adapted into this release package. The exact unavailable boundary is therefore explicit: Draft 3 establishes behavioral underdetermination among declared test doubles and a phase-sensitive difference in their local implementations. It does not establish that the complete Rao-Ballard or Friston process theories lack phase variables, cannot be extended to receiver-relative timing, or lose to SAN as whole programs. A later benchmark must preregister faithful, capacity-matched implementations with their authors’ equations and inference procedures intact.

12. Three distinct conclusions

The paper recognizes three evidential levels:

1. **Narrative-architecture equivalence:** the ordered roles and feedback loop match under the CIS. Draft 2 retains and sharpens this bounded claim.

2. **Operational equivalence:** observable transitions are preserved in both directions under a formal translation. Draft 2 certifies weak behavioral equivalence for the declared finite grammars, 970 source terms, 3,880 term-context pairs, and four contexts. It does not certify a whole tradition or full biophysical SAN model.
3. **Empirical identity:** the mapped neural process is the mechanism that generates the reported experience. Draft 1 proposes tests; it does not establish identity.

A fourth claim-metaphysical identity between SAN and a religious doctrine-is outside the scope of the method.

13. Falsifiable experiments

13.1 Predictable background versus informative deviation

Use a learned sensory sequence with manipulated predictability. Estimate q_t , event surprisal, phase relationships, population activity, and downstream discrimination. TCT predicts that receiver-specific residuals and their causal influence will outperform raw power or global synchrony alone in predicting reported content.

Falsifier: raw synchrony or power fully accounts for content-specific performance after residual and receiver-state terms are controlled.

13.2 Synchrony-differentiation dynamic range

Compare wakefulness, task engagement, NREM sleep, anesthesia, focal seizure evolution, and postictal states using matched estimators. Test whether successful content access occupies a regime with both coordination and structured differentiation.

Falsifier: consciousness and content are monotonic functions of global synchrony across the tested conditions, with no contribution from structured differences.

13.3 Hallucination and source attribution

Compare externally driven perception, voluntary imagery, dreaming, sensory-deprivation hallucination, and clinically characterized hallucination. Decode content and estimate attribution α_t . TCT predicts overlapping content geometry with separable evidence-weighting and self-model states.

Falsifier: internally and externally generated contents require wholly unrelated representational dynamics, with no common receiver-relative rendering operation.

13.4 Content-minimal awareness

Use trained participants, preregistered phenomenological interviews, physiological monitoring, and independent state classification. The target is not proof of “pure consciousness,” but whether participants can report reduced differentiated content while maintaining continuity and later memory.

Falsifier: every reliable report of content-minimal awareness is better explained by sleep, microsleep, amnesia, demand characteristics, or unmeasured ordinary content.

13.5 Intrinsic rendering perturbation

Perturb a candidate rendering state while holding sensory input as constant as possible. Measure content discrimination, action, reafference, memory, and future reference-state changes. Compare against perturbing an external display buffer that is not inside the agent’s own control loop.

Falsifier: candidate neural rendering states can be removed without measurable effect on discrimination, action, memory, or future processing, while the attributed content remains intact.

13.6 Formal encoding benchmark

Draft 2 implements finite transition systems for L_P and L_S , an explicit translation, semantic name substitutions, binary choice, binary parallel composition, silent target transitions, weak observable moves, weak success, divergence detection, and four interleaving contexts. The next benchmark must add synchronized communication, channel restriction, name passing, recursion, larger term depths, and independently authored adversarial contexts.

Falsifier: no nontrivial translation survives the five criteria without erasing the distinctions that motivated the comparison.

13.7 Discriminative comparator and ablation benchmark

The bounded translation must also be tested against mechanisms that can mimic a generic five-stage loop. Compare a passive framebuffer with no self-use or update, a pure synchrony oscillator with no typed difference, a feedforward classifier with no return path, an unstructured noisy recurrent network, a standard predictive controller or active-inference loop, and the complete SAN receive-transform-use-act-update chain. Then ablate receiver-relative typing, structured differentiation, intrinsic observation, embodiment, reafference, and memory update one at a time.

The protected result is not that only SAN can implement a feedback loop. It is that the declared SAN signature cluster identifies which functional invariants are required for this particular translation and which comparators share only a broader narrative architecture.

Falsifier: generic recurrent or predictive systems pass the same protected observables after SAN-specific invariants are removed, leaving no discriminative contribution beyond ordinary feedback control.

Draft 3 execution. The deterministic suite produced the following results:

Declared mechanism	Content / action accuracy	Return / memory sensitivity	Phase-flip drop	Disposition
Passive framebuffer	0.50 / 0.00	0.00 / 0.00	0.00	no intrinsic use
Tonic-only coordinator	0.50 / 0.50	2.00 / 2.00	0.00	no differentiated content
Phasic-only detector	0.50 / 0.50	2.00 / 2.00	0.00	context shift defeats content
Pure synchrony oscillator	0.50 / 0.50	0.00 / 0.00	0.00	coordination without typed content
Unstructured dispersion	0.50 / 0.50	2.00 / 2.00	0.00	novelty without information
Feedforward typed classifier	1.00 / 1.00	0.00 / 0.00	0.00	content and action, no return / update
Abstract predictive controller	1.00 / 1.00	2.00 / 2.00	0.00	generic behavioral pass

Declared mechanism	Content / action accuracy	Return / memory sensitivity	Phase-flip drop	Disposition
Active-inference-like controller	1.00 / 1.00	2.00 / 2.00	0.00	generic behavioral pass
SAN tonic-phasic chain	1.00 / 1.00	2.00 / 2.00	1.00	generic and phase-sensitive pass

The seven ablations localize the failure rather than merely lowering one aggregate score:

Single ablation	Protected observable that fails
Remove tonic reference	content, conditional information, action
Remove receiver-relative typing	content, conditional information, action
Remove structured difference	content, conditional information, action
Remove intrinsic observation	action and self-use
Remove embodiment	action and self-use
Remove reafference	return and memory
Remove memory update	memory only

The result sharpens the falsifier. The abstract predictive and active-inference-like controls already show that a generic behavioral loop can reproduce the behavioral score vector. TCT therefore survives this benchmark only as a more specific implementation hypothesis: an empirical study must measure a receiver-relative physical relation and show selective sensitivity to an internal phase perturbation, not merely observe accurate prediction, recurrent feedback, or successful action.

13.8 Receiver-relative phase measurement and intervention

An organism can make the same correct choice through different internal mechanisms. Accurate behavior, recurrent feedback, and successful prediction therefore do not reveal whether a receiver-relative phase relation contributes to the computation. The missing experiment must change that internal relation while keeping the sensory event, stimulation dose, reward structure, and task demands as similar as possible.

The established methods come first. Intracranial recording can estimate sender and receiver phase, evoked responses, spike-field locking, and directed interactions. Closed-loop stimulation can deliver the same charge-balanced pulse at a selected phase of an ongoing rhythm. Kragel and colleagues (2025) demonstrated this feasibility in human cortico-hippocampal networks: lateral-temporal stimulation locked to hippocampal theta increased theta activity and later connectivity relative to approximately rhythm-matched but phase-blind stimulation. That result supports the intervention technology; it does not test TCT, PWD, intrinsic rendering, or consciousness.

In plain language, the proposed study first learns the phase relation normally associated with a content transition for a particular sender, receiver, and context. It then delivers matched perturbations at the predicted relation, at a phase-shifted relation, without phase locking, and under sham. The study asks whether the perturbation changes the receiving circuit, decoded content, action, returned sensory consequence, and later memory in the receiver-specific direction predicted before intervention. Only after that operation is specified is it called a **receiver-relative phase intervention**.

For sender s , receiver r , conduction delay τ_{sr} , context z_t , and history h_t , define the measured residual phase relation

$$\delta_{sr}(t) = \text{wrap}[\phi_s(t - \tau_{sr}) - \phi_r(t) - \hat{\mu}_{sr}(h_t, z_t)]. \quad (26)$$

Here $\hat{\mu}_{sr}$ is estimated only from training data. It is not a universal preferred phase. The closed-loop pulse has fixed waveform and charge and is triggered within a preregistered tolerance around offset κ :

$$\begin{aligned} u_{\kappa}(t) &= u_0 \mathbf{1}(|\text{wrap}[\delta_{sr}(t) - \kappa]| \leq \varepsilon), \\ \kappa &\in \{0, \pi/2, \pi\}. \end{aligned} \quad (27)$$

Pulse count, mean interval, charge, anatomical site, and task timing are matched across phase-targeted conditions. Separate phase-blind and sham conditions control for rhythmic stimulation and nonspecific intervention effects. A nested held-out comparison tests whether the receiver-relative term adds information beyond rate, power, evoked amplitude, task state, and abstract prediction error:

$$\Delta_{CV} = \mathcal{S}_{\text{test}}(M_{\text{base}} + \delta_{sr} + \delta_{sr} \times r + \delta_{sr} \times z_t) - \mathcal{S}_{\text{test}}(M_{\text{base}}). \quad (28)$$

$\mathcal{S}_{\text{test}}$ is a preregistered held-out score, not a training fit. M_{base} includes stimulus identity, behavior, firing rate, spectral power, arousal, movement, and a source-bound predictive or active-inference state estimate where applicable.

Stage A: observational localization. Use human intracranial data collected for clinical reasons, or a suitable animal model, to estimate $\hat{\mu}_{sr}$, delay, receiver identity, and context dependence. All choices of frequency band, channel, time window, directionality estimator, artifact rejection, and decoding endpoint are fixed before held-out evaluation. Candidate outcomes include receiver evoked response, spike-field locking, directed transfer, content decoding, action latency or accuracy, reafferent response, and delayed memory update.

Stage B: causal intervention. Randomize phase-targeted, shifted-phase, phase-blind, and sham trials under blinded analysis. The primary test is not a main effect of synchrony. It is a preregistered sender-by-receiver-by-context interaction: the phase offset predicted for one receiver and context should not be treated as universally optimal. A rescue condition returns stimulation to the predicted offset after the shifted-phase disruption.

Support criterion. TCT receives mechanism-level support only if (1) $\Delta_{CV} > 0$ on untouched data; (2) the matched phase intervention selectively changes the predicted receiver and downstream causal chain; (3) the shifted-phase condition impairs or redirects that chain; (4) the preregistered phase rescue restores it; and (5) mediation remains after power, rate, arousal, movement, evoked-response, and multiple-comparison controls.

Failure conditions. The proposed discriminator fails if the receiver-relative term adds no held-out value; if phase-blind stimulation produces the same result; if effects follow power, amplitude, arousal, or movement rather than receiver identity; if the predicted interaction does not replicate; if a capacity-matched source-faithful alternative predicts the same internal result without the TCT variable; or if stimulation changes the measurement artifact but not the causal chain. A failure narrows the phase-typed implementation even if behavior remains accurately predicted.

The intervention requires independent ethics, clinical, safety, and preregistration review. This paper specifies a test and does not authorize stimulation, recruitment, diagnosis, or treatment.

14. Relationship to adjacent theories

14.1 Theory-level boundary

Predictive-processing accounts supply a natural language for history-conditioned expectations and residual error (Friston, 2010; Alamia & VanRullen, 2019), but prediction error is not identical to PWD. PWD emphasizes receiver-relative physical transformations, phase relations, multiscale propagation, and the rendering role of the differentiated pattern. Global-workspace accounts emphasize availability; integrated-information approaches emphasize causal organization; recurrent-processing accounts emphasize feedback; active-inference approaches couple perception and action. TCT borrows none of their conclusions merely by using adjacent vocabulary. Its distinctive proposed cluster is:

- tonic reference
 - typed phase-wave difference
 - intrinsic recurrent rendering
 - embodied action / memory update.
- (29)

The comparison with contemplative traditions contributes a second distinction: the absence of a homuncular viewer can be represented mechanically by causal self-access. “The field experiences itself” is thus interpreted as the organized field-cell system producing, detecting, remembering, and changing its own state—not as the claim that any electromagnetic field is conscious.

14.2 Adjacent SAN-paper nonduplication

This paper owns a narrow conjunction: a source-bounded ground / form translation, the tonic-reference versus receiver-relative-difference mechanism, formal encoding criteria, and the proposed phase discriminator. Adjacent papers remain necessary because they own different questions:

Adjacent work	Its primary scope	this paper boundary
<i>The Field-Cell Self, The Field-Cell Self</i>	Distributed biological identity and the observer as a field-cell process	this paper uses the no-homunculus consequence but does not rederive the self theory.
<i>Tomographic Coherence-Based Neural Rendering, TCNR</i>	Public-data neural reconstruction and phase-structured decoding	this paper proposes a mechanism discriminator, not the TCNR reconstruction program.
<i>Phase-Wave Differentials, PWD</i> mathematical analysis	General receiver-relative PWD formalization and a preserved grouped negative benchmark	this paper uses a restricted residual and does not erase <i>Phase-Wave Differentials</i> ’s adverse result.
<i>Choice Without a Central Chooser, Choice Without a Central Chooser</i>	Distributed route selection and action choice	this paper includes action as one translated stage, not a complete choice theory.
<i>Embodied Neural Rendering in Self-Aware Networks, embodied neural rendering</i>	Vision, body maps, sensorimotor loops, and the Gamma Consideration Sandwich	this paper is not a visual-system or embodiment survey.
<i>Phase-Wave Differential Calculus, PWD</i> calculus	Neural tuning, variability, and the Taylor-polynomial sequence	this paper does not repeat that calculus or its token construction.
<i>The Coherence-Dispersion Sculpting Hypothesis, coherence-dispersion sculpting</i>	Altered states, coherence, dispersion, plasticity, and qualia dynamic range	this paper owns the comparative-philosophy translation and its formal audit.
<i>Registering an Embodied World to Brain Dynamics, cross-modal 3D registration</i>	External 3D capture, observer-relative XR, and a scene-to-brain experiment	this paper does not claim 3D registration or volumetric reconstruction.

Adjacent work	Its primary scope	this paper boundary
<i>Self-Aware Networks: A Systems Whitepaper</i> , formal SAN whitepaper	System-wide SAN architecture	this paper is a focused companion and cannot substitute for the complete whitepaper.
<i>Measuring Candidate Machine-Consciousness Mechanisms in Self-Aware Networks</i> , machine-consciousness measurement	Adversarial machine indicators and perturbational validation	this paper does not classify machines as conscious.
<i>Coherence-Weighted Electrical Drive in Choice and Motor Realization</i> , electrical valence and motor choice	Coherence-weighted effective drive and motor realization	this paper uses action and return without asserting the electrical-valence mechanism.
<i>How a Brain-Body Network Detects Its Own Emotional State Changes</i> , neuroaffective self-registration	Emotion, interoception, and neurofeedback genealogy	this paper does not provide an emotion model.
<i>Coincidence as a Bit of Information</i>	Coincidence, temporal summation, and action recruitment	this paper does not claim ownership of the coincidence-information theory.

No inspected adjacent manuscript combines the same source-specific comparative map, finite formal certificates, adverse comparator result, and receiver-relative phase-intervention design. That is the nonduplication basis for retaining this paper as a separate paper. It is a corpus audit result, not a priority claim against outside authors.

15. Limitations and nonclaims

1. No oscillatory frequency band is identified as the universal carrier of tonic awareness.
2. High synchrony is not equated with consciousness, and dispersion is not equated with information.
3. Low information means low conditional surprisal or entropy rate under a declared model, not physical emptiness.
4. The five-role interface has a machine-checked graph-isomorphism certificate, and the bounded grammars have a machine-checked weak-behavioral certificate. Neither result is a full category-theoretic or π -calculus proof for unbounded languages, name passing, actual neural tissue, or whole traditions.
5. The mapping does not erase the doctrinal differences among Advaita, Madhyamaka, Zen, Western theology, and contemporary phenomenology.
6. A first-person report is evidence about experience and report, not a direct measurement of neural mechanism.
7. A successful functional encoding would not by itself solve the explanatory gap or establish machine consciousness.
8. The June-August 2022 Git history establishes Micah Blumberg's public genealogy for this synthesis; the internally dated 2011-2012 records establish authored ancestry only, and none of these dates show that all individual atoms lacked earlier precursors.
9. The predictive and active-inference controls in the executable suite are local test doubles. They are bound to independently authored definitions in Section 11.9, but no author-supplied, source-faithful implementation was adapted to the this paper interface.
10. The receiver-relative phase experiment is a preregisterable proposal. Kragel et al. establish feasibility for phase-locked human stimulation, not evidence for TCT, PWD, intrinsic rendering, or a tonic ground of awareness.

16. Conclusion

The ancient-to-machine bridge can be made more rigorous than analogy, but less sweeping than metaphysical proof. The shared narrative architecture is strong enough to formalize: continuity or ground, arising difference, discernment, action, and consequence map onto tonic reference, PWD, learned tuning, embodied control, and recursive update. The Causal Interface Sequence preserves the direction of the story. The bounded five-role model certifies exact labeled-graph isomorphism. Draft 2 additionally certifies weak behavioral equivalence across the declared finite grammars, semantic substitutions, silent transitions, and four contexts, while eight adversarial mutations fail closed. Category theory states what a larger generalization should preserve; pi-calculus exposes richer interactions and observable traces still outside the bounded proof.

The principal scientific proposal is that conscious neural rendering depends neither on synchronization alone nor on difference alone. It depends on structured, receiver-readable difference within a sufficiently coordinated, recurrent, embodied, and self-updating field. That claim explains why a tonic canvas can be experientially quiet yet mechanically active, why “formlessness” need not mean absence of structure, and why ordinary computer graphics and intrinsic neural rendering are not equivalent merely because both can encode a scene.

The result is a bounded mechanical accounting of a longstanding philosophical motif and a concrete experimental program. Narrative correspondence, finite-interface graph isomorphism, and bounded weak operational equivalence are presently certified at their declared scopes. At the Draft 2 boundary, discriminative comparator tests, whole-language operational equivalence, whole-tradition isomorphism, and empirical neural identity remained open proof obligations. Draft 3 closes only the bounded synthetic comparator test; whole-language equivalence, whole-tradition isomorphism, neural implementation, and phenomenal identity remain open.

Draft 3 closes the bounded synthetic comparator obligation but narrows the interpretation at the same time. The tonic reference, typed phasic difference, intrinsic self-use, embodiment, refference, and memory update are jointly necessary in the declared stress test, yet abstract predictive and active-inference-like controllers reproduce the generic behavioral chain. The remaining empirical question is therefore internal mechanism identification: does a measured receiver-relative phase relation make a selective causal contribution beyond an abstract state estimate? That question can fail, and the synthetic phase-flip result is only a design for asking it.

Draft 4 supplies the missing route from synthetic discrimination to a real test. It does not predict that one frequency, one phase, or more synchrony is always better. It predicts a receiver- and context-specific relation whose contribution must be measured on held-out data and changed causally under matched interventions. The same draft also prevents the philosophical bridge from becoming a false merger: Tillich, the *Mandukya Upanisad*, Advaita, Madhyamaka, Dogen’s Zen, and modern content-minimal-awareness research remain separate source families. The final-preprint claim is consequently precise: TCT is a bounded, formally disciplined, source-traceable, and falsifiable translation proposal, not an established neural identity or a proof of any metaphysical tradition.

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Appendix A. SAN genealogy anchors and date grades

The table keeps internal authorship dates separate from public fixation. A later Git commit can establish that text was public by that commit; it cannot independently prove the earlier date written inside the note.

Internal or owner date	Public Git fixation	Source	Relevant atom	Commit	Evidence grade
2011-05-13	2022-06-08	original a0348z ctp.r.txt, now a0348z.md	distributed scene pixels, learned statistical relations, prediction, continuously updated multisensory picture awareness as prediction / readiness; empty luminosity; embodiment and movement calculate current-versus- desired neural difference and transmit the difference	63d76b0b1b5ea1cfbfdd0699 4ddd180c2e8ee5ea	owner-dated ancestry; public by 2022-06-08
2012-08-20	2022-06-10	a0003z.md	phasic spikes evaluated relative to an oscillator's tonic firing pattern	7737b4f69f0de736956f6532db bfc868072dcd9f	owner-dated ancestry; public by 2022-06-10
attributed 2017	unresolved	nlp1.md	coherence for binding versus differentiation for distinction; oscillator creates its own reference paper	source-gated	transcript ancestry; exact public 2017 date not yet certified
-	2022-06-08	original a0245z ctp.r.txt, now a0245z.md	intrinsic rendering, no separate viewer, near-future prediction for coordinated action	a1180bfb254bb7b4329a447f 71d9d6df45e78bce	public mechanism anchor
-	2022-06-08	original a0142z ctp.r.txt, now a0142z.md	exact tonic-canvas / phasic-ink formulation and common- versus-rare context	37da575ba33f85cef4239295be 161066d8d39f4c	public mechanism anchor
-	2022-06-08	original a0135z ctp.r.txt, now a0135z.md	spiritual ground-of-being motif mapped to tonic oscillation / restful sensory condition	37da575ba33f85cef4239295be 161066d8d39f4c	public mechanism anchor
-	2022-06-11	a0245z.md	tonic oscillatory activity as ground of awareness from which content arises	df6e60216b38a2475a7819cfdb 2baf2d3e3eb2e	public signature-cluster anchor
-	2022-08-04	a0003z.md	firefly synchronization; tonic common / low-information versus phasic rare / high- information; multiscale bidirectionality	f739086d642ecf3a04bc259f94 09a3f3be0d633c	public contemplative join
-	2022-08-21	a0004z.md	tonic oscillation as canvas of consciousness	6ada31d7eca84f8c4666440cb4 c9eb5638f27963	public contemplative join
-	2022-08-24	WhitepaperNotes.md	redness as PWD relative to tonic baseline; ground of awareness / being as readiness state	b56ae48e8abac67f3c3e89ee64 1c360f34b39f02	public synthesis anchor
-	2022-08-24	whitepaperdraft2.md	canvas / paint / qualia and distributed oscillating-group observer formulation	b56ae48e8abac67f3c3e89ee64 1c360f34b39f02	public synthesis anchor
-	2022-08-24	a0004z.md	differentiated and valenced forms arising within formless tonic awareness	b56ae48e8abac67f3c3e89ee64 1c360f34b39f02	public qualia / PWD anchor
-	2022-08-26	00changelog.md		a5f25729d14bc8805cc6993fe9 9156bf6351920	public synthesis anchor
-	2022-09-04	a0004z.md		29369061baee78f8974e30a8 e48f1bc95ae91fa2	public ground / form anchor

Appendix B. Formal proof obligations

Obligation	Required artifact	Current status
Source grammar	Typed syntax for bounded phenomenological states and transitions	executable depth-two grammar plus complete five-stage trace family; 970 normalized terms; unbounded grammar open
Target semantics	SAN transition terms with visible and silent steps, choice, parallelism, state, and updates	bounded target grammar implemented; channels, synchronization, restriction, recursion, and name passing open
CIS preservation	Machine-readable atom / edge correspondence	finite five-role graph pass plus complete five-stage trace-family pass
Compositionality	Translation contexts for each source constructor	bounded pass for zero, success, prefix, choice, and parallel over 970 terms
Name invariance	Semantic substitution and false-friend tests	injective semantic content / value / action / memory substitutions pass; cross- tradition false-friend suite open

Obligation	Required artifact	Current status
Operational completeness	Each source reduction weakly mimicked by target	pass across 3,880 term-context pairs
Operational soundness	Each claimed target-visible trace corresponds to source	pass across 3,880 term-context pairs
Divergence reflection	Detect target-only silent livelock	pass; baseline has no reachable tau cycle and tau-divergence mutation fails
Success sensitiveness	Bidirectional preservation of weak success	pass in four contexts; premature-success mutation fails
Category mapping	Partial functor with declared domain and composition tests	finite-interface identity / composition pass; general categorical proof open
Discriminative comparison	Comparator and ablation benchmark for protected SAN invariants	bounded synthetic pass: nine comparators, seven ablations, 768 trials; behavioral nonuniqueness retained; empirical mechanism test open
Comparator provenance	Independently authored definitions and exact implementation boundary	Rao-Ballard and Friston definitions pinned; local controls disclosed as test doubles; source-faithful implementations remain open
Empirical bridge	Preregistered estimators, perturbations, nulls, and falsifiers	two-stage receiver-relative observation / intervention protocol specified in Section 13.8; execution remains open

Appendix C. Executable artifact register

Artifact	Role	Current disposition
model/bounded-interface-model.json	Draft 1 five-role graph certificate	preserved rollback baseline
validation/bounded-interface-proof-receipt.json	Draft 1 exact graph-isomorphism receipt	preserved rollback baseline
model/bounded-process-encoding-v2.json	Draft 2 grammar, label map, bounds, contexts, and claim ceiling	active bounded model
tools/validate_bounded_process_encoding.py	exhaustive weak-behavior, Gorla-style, divergence, and mutation validator	passes
tools/test_bounded_process_encoding.py	independently invocable unit-test layer	three test groups pass
model/weak-bisimulation-witness-v2.json	counts, representative terms, relations, and mutation dispositions	generated witness
validation/bounded-process-encoding-v2-initial-failure.json	preserved pre-correction weak-success failure	retained negative baseline
validation/bounded-process-encoding-v2-receipt.json	final bounded proof receipt	pass: 970 terms, four contexts, 3,880 pairs, eight mutations

model/tonic-phasic-comparator-suite-v3.json | Draft 3 trial design, comparator definitions, thresholds, and claim ceiling | active bounded comparator model |
tools/run_tonic_phasic_comparator_suite.py | deterministic comparator, intervention, and ablation runner | passes |
tools/test_tonic_phasic_comparator_suite.py | independent unit and mutation tests | five test groups pass |
model/tonic-phasic-comparator-results-v3.json | complete per-mechanism and per-ablation result surface | generated pass |
validation/tonic-phasic-comparator-suite-v3-receipt.json | bounded execution receipt | pass: 768 trials, nine comparators, seven ablations |
sources/comparator-tradition-register-v1.json | Tradition-specific and comparator source identities, definitions, and boundaries | active Draft 4 source register |

protocol/receiver-relative-phase-intervention-v1.json | Machine-readable observational and causal study contract | specified; not executed or authorized |
CLAIM-SOURCE-MATRIX-DRAFT4.md | Draft 4 claim, source, genealogy, and failure-condition layer | active final-preprint matrix |
reviews/COMPARATIVE-PHILOSOPHY-COMPARATOR-AND-PHASE-PROTOCOL-AUDIT-20260726.md | source and protocol audit | active |
reviews/ADJACENT-PAPER-NONDUPLICATION-AND-LOSSLESSNESS-AUDIT-20260726.md | corpus boundary and Draft 3 preservation audit | active |

No executable artifact authorizes public release. The proof ceiling is the declared finite language and context family; publication, empirical, whole-language, and metaphysical conclusions require separate evidence and author approval.